

LESSON-5

EQUILIBRIUM

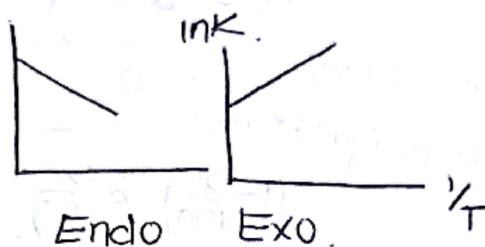
i. Chemical Equilibrium.

ii. Ionic Equilibrium.

$$\text{Degree of dissociation} = \frac{\text{No. of moles dissociated}}{\text{Total No. of moles}}$$

$$\alpha = \frac{D-d}{(n-1)d}$$

$$\alpha = \left(\frac{M_t - M_0}{M_0} \right) \times \frac{1}{(n-1)}$$



where.

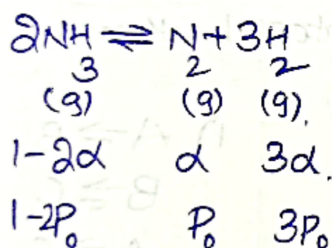
$\alpha \rightarrow$ Degree of dissociation.

$D \rightarrow$ Vapour density ... ($\because D \propto \frac{1}{V}$)

(i) K_p & K_c has no units ...

(Molecular Mass = 2.VD)

(ii) K_c Calculation:



$$P_T = 1-2P_0 + P_0 + 3P_0$$

$$P_T = 1+2P_0$$

$$K_c = \frac{K_f}{K_b}$$

"Gas constant:"

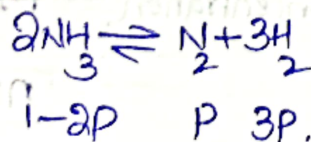
$$0.0821 \text{ Latm/mol-K}$$

$$8.314 \text{ J/mol-K}$$

$$\text{m}^3 \text{ Pa/mol-K}$$

$$2 \text{ cal/mol-K}$$

(iii) K_p Calculation:



$$K_p = \frac{[N_2][H_2]^3}{[NH_3]^2} = \frac{\left(\frac{P}{1+2P} \right) P_T \times \left(\frac{3P}{1+2P} \right)^3 P_T^3}{\left(\frac{1-P}{1+2P} \right)^2 P_T^2}$$

$$\text{Partial Pressure} = X \cdot P_T$$

= (Mole fraction) (Total Pressure) ...

$$PV = nRT \dots \left[d = \frac{M}{V} \right] \dots$$

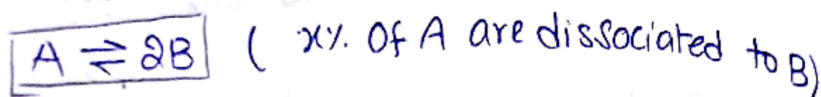
Relation B/w K_p & K_c :

$$K_p = K_c (RT)^{\Delta n}$$

where

$\Delta n = \text{NO. of Moles of Product} - \text{NO. of Moles of Reactant}$

Writing Moles at equilibrium:



Initial Moles 1 0

Moles at equilibrium

$$\left(1 - \frac{x}{100}\right) \quad 2\left(\frac{x}{100}\right) \rightarrow (\text{NO. of Moles}) \cdot x$$

$$\log \frac{K_2}{K_1} = \frac{\Delta H}{2.303R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right] \rightarrow \text{initial Pressure}$$

$$\Delta G = -2.303RT \log K \dots$$

Equilibrium:

rate of forward = rate of backward.

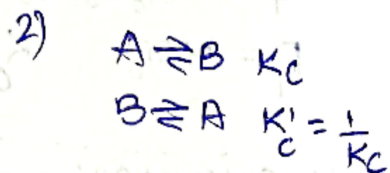
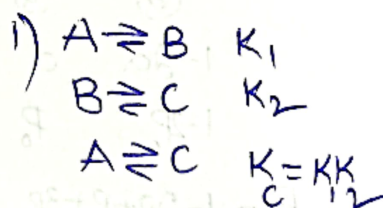
Dynamic in Nature.

$$\Delta G = 0.$$

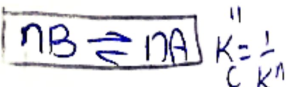
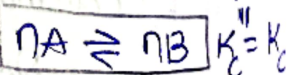
Note:

$$K_c = \left(\frac{\text{Mole}}{\text{Lt}} \right)^{\Delta n}$$

$$K_p = (\text{atm})^{\Delta n}$$



Active Mass = Molar Concentration.



$$\Delta G = \Delta G^\circ + RT \ln Q.$$

$$\Delta G^\circ = -RT \ln K, -2.303RT \log K.$$

Prediction:

i. $Q_c < K_c$: Forward.

ii. $Q_c > K_c$: Backward.

iii. $Q_c = K_c$: Equilibrium.

For $aA + bB \rightleftharpoons cC + dD$.

$$Q_c = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

ii. Acids/Bases:

TOPICS:

1. Arrhenius Concept

2. Bronsted Lowry Theory Bronsted Acid: Proton Donor

3. Lewis Concept: Bronsted Base: Proton Acceptor

→ Lewis Acid: e^- Pair Acceptor.

→ Lewis Base: e^- Pair Donor.

Neutralization: Acid + Base $\rightleftharpoons$ Salt + Water.

4. Types of Solvents:

→ Protophilic (Accept Proton)

→ Protogenic (Produce Proton)

→ Amphiprotic/Amphoteric Accept/Produce.

→ Aprotic: (Neither Donate nor Accept)

Ionic Product of Water (K_w)

- $p_{K_w} = 14$

- $K_w = 1 \times 10^{-14} \text{ moles/Lit}^2$

- $T \propto K_w$

$$pH = -\log H^+$$

$$pOH = -\log OH^-$$

$$pH + pOH = p_{K_w} = 14$$

- Pure water, Neutral sol $\Rightarrow pH = pOH = 7$

$$pH = \text{Initial } pH + \log \frac{V_2}{V_1}$$

$$pH = \log \frac{V_2}{V_1}$$

Two
S.A

$$H^+ = \frac{N_1 V_1 + N_2 V_2}{V_1 + V_2} = \frac{M_1 V_1 n_{\text{factor}} + M_2 V_2 n_{\text{factor}}}{V_1 + V_2}$$

Two:
S.A

$$H^+ = \frac{N_1 V_1 - N_2 V_2}{V_1 + V_2}$$

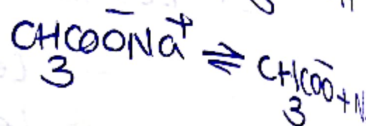
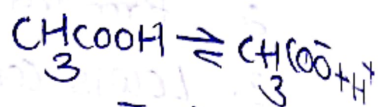
Limited
Amount
of SB.

* For Amphiprotic Solvents:

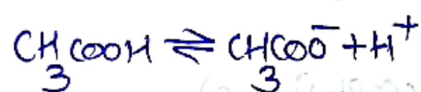
$$pH = \frac{pK_{a1} + pK_{a2}}{2}$$

Common Ion Effect:

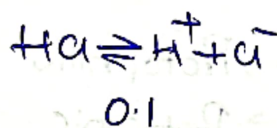
Weak electro - Strong electro.
Common ion.



1. Dissociation Constant of 0.01M of CH_3COOH is 1.8×10^{-5}
then calculate CH_3COO^- Concentration in 0.1M HCl.



$$\begin{array}{ccc} 0.01 & 0 & 0 \\ 0.01-x & x & x+0.1 \end{array}$$



$$0.1$$

Add 0.1 as common ion...

Relative Strength:

$$\frac{\text{Strength of Acid}_1}{\text{Strength of Acid}_2} = \sqrt{\frac{K_{a1}}{K_{a2}}}$$

- No. of H^+ release is that
Basic [Mono, di, tri, poly]

- Any Salt is Strong Electrolyte.

pH Calculations:

1. Strong Acid / Strong Base:

$$\begin{array}{l} H^+ < 10^{-6} \text{ + need to add } 10^{-7} \text{ from water.} \\ H^+ \geq 10^{-6} \end{array}$$

$$\text{Ex: } 10^{-4} \text{M HCl.}$$

$$pH = -\log 10^{-4}$$

$$pH = 4$$

$$\text{Ex: } 10^{-8} \text{M HCl.}$$

sol: need to add 10^{-7} as Acid pH is less than 7 Only 10^{-8} to 10^{-7}

$$pH = \log [10^{-8} + 10^{-7}]$$

$$\begin{array}{l} \log 2 = 0.3 \\ \log 3 = 0.47 \quad \log 4 = 0.6 \\ \log 5 = 0.7 \\ \log 6 = 0.77 \\ \log 7 = 0.84 \\ \log 8 = 0.9 \\ \log 9 = 0.94 \end{array}$$

2. PH of two SA (or) SB: M_1, M_2 : Molarity...
 N_1, N_2 : Normality
 V_1, V_2 : Volume.

$$H^+ = \frac{N_1 V_1 + N_2 V_2}{V_1 + V_2}$$

$$H^+ = \frac{M_1 V_1 n_{\text{factor}} + M_2 V_2 n_{\text{factor}}}{V_1 + V_2}$$

3. PH of SA + O.B:

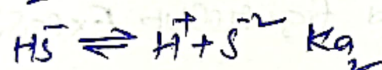
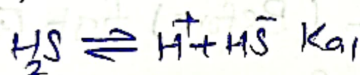
$$\frac{H^+}{OH^-} = \frac{N_1 V_1 - N_2 V_2}{V_1 + V_2}$$

4. PH of Monobasic WA (or) WB

$$P_{HWA} = \frac{1}{2}(P_{Ka} - \log C)$$

$$P_{H_{WB}} = \frac{1}{2}(P_{Kb} - \log C)$$

5. PH of Polybasic Acid:



$$P_H = P_{Ka} + \log \frac{\text{Salt}}{\text{Acid}}$$

$$P_H = \frac{1}{2}[P_{K_{a1}} - \log C]$$

↓
 As $P_{K_{a1}}$ is greater than $P_{K_{a2}}$.

6. $H^+ = \sqrt{K_{a1} C_1 + K_{a2} C_2}$ (C_1, C_2 Conc After Mix)

PH of Mix of two WA or two WB:

7. Buffer Capacity: $\frac{\text{no. of Moles of an Acid added per lit sol.}}{pH \text{ change.}}$

• Compound having Low K_{sp} Precipitates first...
 Ionic Product $> K_{sp}$.

$$\text{Normality} = \frac{\text{Sp. gr} \times \% \times 10}{\text{Eq. wt}}$$

• Solubility of Sparingly Soluble Salt $\Rightarrow K = K_{eq} \times K_{sp}$.

• Write the reaction when any type of reactant is given..

$$K = K_1 \cdot K_2$$

$$K = \frac{C \alpha^2}{1 - \alpha}$$

Small α change: $1 - \alpha \approx 1$

$$K = C \alpha^2$$

$$H^+ = \sqrt{K_a \times C}$$

$$\alpha = \frac{\text{No. of Moles of dissociated into ions}}{\text{Total Moles}}$$

($\because K_a$ dissociation Constant)

$$K_b = C\alpha^2$$

$$K_a = C\alpha^2$$

$$L \text{ atom/Mol} \quad 0.021$$

$$J/\text{Mol} \quad 8.314$$

$$C_{\text{cell}}/\text{Mol} \quad 2$$

$$m^3 \text{ Pa/Mol K} \quad 8.314$$

$$L\text{-Torr Mol} \quad 62.36$$

- When two comp Add (Acid, Base) then Determine Limiting reactant (Before) first Find Both Reactants Moles and figure out Excess Part in it...

1. Moles of Both reactants

2. Figure out Limiting Part.

Energy